

Sébastien Marchand

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Research interests

Environmental economics, agricultural economics, climate economics, institutional economics, and development economics.

Methods: applied econometrics (impact evaluation, spatial econometrics, production-frontier analysis), text and sentiment analysis, and field data.

My research is organised around four main areas:

- **Area 1 – Agricultural economics.** Productivity, sustainable practices, and ecosystem services (China, Brazil, Cambodia).
- **Area 2 – Climate economics.** Vulnerability, exposure to extreme events, and climate finance (Vietnam and developing countries).
- **Area 3 – Media economics and environmental perceptions.** Emotional tone in the media and air quality (China).
- **Area 4 – Environmental economics (other topics).** Trade restrictions, deforestation, and agri-environmental schemes.

Education

- **2022 – Habilitation to Supervise Research (HDR),** Université Clermont Auvergne.
 - Title: *The sustainability of agricultural production systems in developing countries* [in French].
 - Academic sponsor: Catherine Araujo Bonjean (CNRS Research Fellow, CERDI).
 - Committee: T. Bernard (Professor, Bordeaux School of Economics), S. Blancard (Professor, Agrosup Dijon, CESAER), R. Chakir (Research Director, INRAE, PSAE), and V. Dequiedt (Professor, UCA, CERDI).
- **2011 – PhD in Economics,** highest honours with jury congratulations, CERDI, Université d'Auvergne.
 - Title: *Institutions and deforestation in developing countries*.
 - Supervisors: Pascale Combes Motel and Florent Garnier.
 - Committee: C. Dugas de la Boissonny (Emeritus Professor, Nancy 2), G. Duchêne (Professor, Université Paris-Est Créteil), L. Hotte (Professor, University of Ottawa), and J. Férès (IPEA, Rio de Janeiro).
- **2008 – Research Master's degree in Development Economics,** Université d'Auvergne (Merit).

- **2008** – Professional Master's degree in Project Analysis, Université d'Auvergne (Merit).

Academic positions

- **Since 2013** – Associate Professor of Economics (HDR since 2022), CERDI, Université Clermont Auvergne.
- **2012** – Postdoctoral Researcher, [Rede Amazônia Sustentável](#) project, Federal University of Viçosa, Brazil.
- **2011–2012** – Temporary Teaching and Research Fellow, Université d'Auvergne, Clermont-Ferrand.
- **2008–2011** – Doctoral Research Fellow and Teaching Assistant, Université d'Auvergne.
- **2007** – Research Assistant, Métafort Joint Research Unit (AgroParisTech, INRA, VetAgro Sup).

Scientific and teaching responsibilities

- **Since 2017** – Co-Director of the Master's track *Economics and Territorial Transitions (ETT)*, School of Economics, UCA.
- **Since 2023** – Scientific Coordinator of the IHEDD/FERDI executive training programme *Steering the Ecological Transition* (Tunis, 2024; Clermont-Ferrand, 2025 and 2026).
- **Since 2023** – Coordinator of the Economics Work Package of the ANR-PRC **ECOTERM** project (see *Research grants and funding*).
- **Since 2016** – Coordinator of the *Environment and Economic Policy* module in the GPE Master's programme (UCA/FERDI).
- **2013–2021** – Director of the Bachelor's degree in Economics and Social Administration (approximately 500 students), School of Law, UCA.
- **2013–2017** – Director of the Master's track in Local Economics and Territorial Development, UCA.

Elected and institutional service

- **Since 2026** – Elected member of the Academic and Student Life Council (CFVU), UCA.
- **Since 2024** – Elected member of the School of Economics Council, UCA.
- **Since 2022** – Full member of **CNECA Section 9** (National Commission for Higher Education and Research in Agriculture, Ministry of Agriculture and Food – Economics, Humanities and Social Sciences).
- **2021–2026** – Elected member of the UCA Board of Directors.
- **2017–2020** – Elected member of the CERDI Laboratory Council.

Bibliometric indicators (12 February 2026)

- **15** articles in international peer-reviewed journals (**5 A-ranked**, **6 B-ranked** according to the CNU Section 05 classification, and 4 outside the CNU classification); **8** articles as first or sole author.
- **3** published book chapters; **6** works in progress or under submission (including to *World Development*).
- Google Scholar citations: see [profile](#).

Articles in international peer-reviewed journals

Underlined names indicate current or former PhD students and postdoctoral researchers whom I co-supervise or have co-supervised.

1. **S. Marchand**, D. Cubizol, E. Nasho Ah-Pine, H. Guo (2025). Mass media as an environmental watchdog? Evidence from the war on air pollution in China. *Environment and Development Economics*, 1–25. **[B-ranked]**
2. **S. Marchand**, M. Zerbo (2025). Enhancing employment through log export ban: insights from wood-processing firms in developing countries. *Forest Policy and Economics*, 174, 103491. **[HC CNU – 5-year JIF 4.0]**
3. **S. Marchand**, M. Barro, H. Xiong, H. Guo (2025). Industrial water pollution and farmer adaptation: evidence from rice farming in Jiangsu, China. *Agricultural Economics*, 56(1), 108–123. **[A-ranked]**
4. T. B. Diop, S. Blancard, S. Legras, **S. Marchand**, L. Védrine (2024). Exploring the linkages between farm efficiency, farm environmental performance, and agri-environmental scheme adoption: lessons from France. *Journal of Environmental Management*, 366, 121519. **[B-ranked]**
5. S. Mankai, **S. Marchand**, N. H. Le (2024). Valuing insurance against small probability risks: a meta-analysis. *Journal of Behavioral and Experimental Economics*, 109, 102181. **[B-ranked]**
6. R. Muon, C. Lai, V. Hervé, R. Zaiss, F. Chassagne, E. Bureau-Point, **S. Marchand**, M. Audibert, J. Berger, F. Wieringa *et al.* (2023). Abundance, perceptions and utilizations of termite mounds in Cambodia. *Soil Use and Management*, 39(3), 1172–1184. **[HC CNU – 5-year JIF 4.0]**
7. **S. Marchand**, M. Zerbo (2023). Revisiting the trade restrictions–industrialisation nexus in developing countries: the case of log export ban and wood processing. *Journal of Forest Economics*, 38(2), 195–233. **[B-ranked]**
8. H. Guo, **S. Marchand** (2019). Social interactions and spillover effects in Chinese family farming. *The Journal of Development Studies*, 55(11), 2306–2328. **[A-ranked]**
9. R. D. Garrett, T. A. Gardner, T. F. Morello, **S. Marchand**, J. Barlow, D. Ezzine de Blas, J. Ferreira, A. C. Lees, L. Parry (2017). Explaining the persistence of low income and environmentally degrading land uses in the Brazilian Amazon. *Ecology and Society*, 22(3). **[HC CNU – 5-year JIF 5.3]**
10. **S. Marchand** (2016). The colonial origins of deforestation: an institutional analysis. *Environment and Development Economics*, 21(3), 318–349. **[B-ranked]**
11. A. Sauquet, **S. Marchand**, J. Férès (2014). Protected areas, local governments, and strategic interactions: the case of the ICMS-Ecológico in the Brazilian state of Paraná. *Ecological Economics*, 107, 249–258. **[A-ranked]**
12. **S. Marchand**, H. Guo (2014). The environmental efficiency of non-certified organic farming in China: a case study of paddy rice production. *China Economic Review*, 31, 201–216. **[A-ranked]**
13. T. A. Gardner, J. Ferreira, J. Barlow, A. C. Lees, L. Parry, **S. Marchand** *et al.* (2013). A social and ecological assessment of tropical land-uses at multiple scales: the Sustainable Amazon Network. *Philosophical Transactions of the Royal Society B*, 368(1619). **[HC CNU]**
14. **S. Marchand** (2012). The relationship between technical efficiency in agriculture and deforestation in the Brazilian Amazon. *Ecological Economics*, 77, 166–175. **[A-ranked]**

15. **S. Marchand** (2012). Legal origin, colonial origins and deforestation. *Economics Bulletin*, 32(2), 1653–1670. [B-ranked]

Book chapters

1. M. Goujon, **S. Marchand**, O. Santoni (2026, forthcoming). *The Physical Vulnerability to Climate Change Index: what role can it play in the analysis and evaluation of adaptation policies?* [in French]. In H. Gérardin & M. Clément (eds.), *Development Policies, Actors and Evaluations*, Éditions EME.
2. **S. Marchand**, M. Goujon (2023). *Measuring vulnerability to climate change* [in French]. In F. Cafarelli (ed.), *The Resilience of Territories Exposed to Natural Hazards*, Mare & Martin.
3. M. Goujon, **S. Marchand**, O. Santoni (2023). *A comparative measure of territories' vulnerability to cyclonic activity: the case of Vietnam* [in French]. In R. Bentirou Mathlouthi & A. Pomade (eds.), *Environmental Vulnerabilities: Multidisciplinary Perspectives*, L'Harmattan.

Work in progress and under submission

1. C. Lao, M. Audibert, P. Jouquet, R. Muon, K. Sok, **S. Marchand** (2025). Beyond the surface: when termite mounds affect rice production in Cambodia? *Under submission to World Development*.
2. P. Beaucoral, M. Goujon, **S. Marchand** (2025). No plan, no aid? The effects of National Adaptation Plan implementation on received adaptation aid.
3. S. Chheang, **S. Marchand**. Termite mounds, input allocation, and agricultural misallocation.
4. **S. Marchand**, D. Cubizol, H. Guo. Sentiment analysis and air pollution: evidence from Weibo posts in China.
5. **S. Marchand**, D. Cubizol, H. Guo. Sentiment analysis and air pollution: evidence from industrial firms in China.
6. Y. Diallo, **S. Marchand**, E. Espagne (2019). Impacts of extreme climate events on technical efficiency in Vietnamese agriculture. *CERDI Working Papers 2019/12*; also published in the *AFD Research Papers* series.

Research grants and funding

- **2023–2027 – ECOTERM project** (*ECOsystème services derived from TERmite Mounds in the lower Mekong basin*), ANR-PRC, 48 months.
Coordinator of one of four work packages – allocated work-package budget: €110,000 (approximately 25 % of the total budget), including funding for a postdoctoral position in 2026. A further 25 % of the project budget funds a PhD scholarship that I co-supervise (one-third share).
- **2019–2022** – I-Site CAP 20-25 PhD scholarship (funding for N. H. Le's doctoral dissertation).
- **2011–2015** – *Rede Amazônia Sustentável* project: international research project on sustainable development in the Legal Amazon.
- **2012** – Postdoctoral fellowship, Brazilian Ministry of Science and Technology.
- **2008–2011** – Doctoral fellowship, French Ministry of Higher Education and Research.
- **2008–2011** – Fellowship under the *International Governance, Universalism and Relativism* programme,

French Ministry of Research.

PhD and postdoctoral supervision

Current postdoctoral researcher

- **Sreyphhea Chheang** (since November 2025) – postdoctoral researcher funded by the ANR ECOTERM project; agricultural economics and input misallocation in Cambodia.

Completed PhD dissertations (3)

- **Yoro Diallo** (supervisor: M. Goujon; co-supervision: 50 %) – *Socio-economic impacts of climate change in developing countries* – defended on 6 July 2021 (3 years) – funded by the French Ministry of Higher Education and Research. *Current position*: International Monetary Fund, Young Professionals Program.
- **Mouhamed Zerbo** (supervisor: B. Laporte; co-supervision: 50 %) – *Three essays on the impact of the log export ban policy in developing countries* – defended on 12 December 2023 (4 years) – country-funded scholarship. *Current position*: Temporary Teaching and Research Fellow, University of Limoges.
- **Maimouna Barro** (supervisor: M. Audibert; co-supervision: 50 %) – *Three essays on challenges facing agriculture in developing countries* – defended on 11 January 2024 (5 years) – country-funded scholarship. *Current position*: Temporary Teaching and Research Fellow, UCA.

Current PhD dissertations (2)

- **Pierre Beaucoral** (supervisor: M. Goujon; co-supervision: 50 %) – *Climate and development finance: analysing the determinants and policy levers of climate finance for developing countries* – started October 2022 – funded by the French Ministry of Higher Education and Research.
- **Chanrithy Lao** (supervisor: M. Audibert; co-supervision: one-third, with P. Jouquet, IRD/iEES-Paris) – *Ecosystem services provided by biodiversity in agricultural landscapes in Cambodia* – started November 2023 – funded by ANR (ECOTERM project).

Uncompleted PhD dissertation

- **Ha Ngoc Le** (supervisor: M. Goujon; co-supervision: one-third, with S. Mankai) – I-Site CAP 20-25 scholarship (2019–2022) – discontinued for health reasons.

Scientific service

HDR academic sponsor

- **Lionel Védrine** (INRAE Research Fellow, UMR Territoires) – HDR defended in June 2025.

PhD dissertation reviewer

- **Roxane Sybile Simamindra**, University of Pau and the Adour Region, December 2023 – *An analysis of the socio-economic and environmental opportunities of the circular economy: the case of Africa and Indian Ocean islands*.
- **Badreddine Aboufarasse**, University of Bordeaux, December 2024 – *Essays of the distributional impacts and social acceptability of carbon pricing policy in China*.

Referee for peer-reviewed journals

Agricultural Economics; American Journal of Agricultural Economics; China Economic Review; Ecological Economics; Economic Inquiry; Economics of Governance; Environment and Development Economics; Environment, Development and Sustainability; Frontiers in Ecology and the Environment; Géographie, Économie et Société; International Economics; Journal of Productivity Analysis; Mondes en développement; Revista de Economia e Agronegócio; Revue d'Économie du Développement; Revue d'économie politique; Revue Économique; Scientific Reports; World Development.

Scientific organisation

- 2023 – Member of the organising committee for the **first International Conference on Development Economics (ICDE)**, Clermont-Ferrand.

Research presentations

Conferences (selected)

38th ATM Conference (Bordeaux, 2023); FAERE (Grenoble, 2021); 24th EAERE Conference (Manchester, 2019); *Land use and environmental socio-economic vulnerability*, INRA-CERDI (Clermont-Ferrand, 2018); RAS Symposium (Caraça, Brazil, 2015); AFSE (Lyon, 2014; Aix-en-Provence, 2013); 20th EAERE Conference (Toulouse, 2013); INFER (Orléans, 2013); ISEE (Rio de Janeiro, 2012); ADRES Doctoral Conference (Marseille, 2012); CSAE (Oxford, 2011); CERDI International Conference (Clermont-Ferrand, 2010); NAKRE Research Day (Utrecht, 2010).

Invited seminars

CEE-M (Montpellier, 2022); GREThA (Bordeaux, 2021); CLERMA (Clermont-Ferrand, 2021); Nanjing Audit University – IEF, China (2017); LEO (Orléans, 2011); BETA (Nancy, 2011).

Teaching

Degree programmes (selected)

- **First-year undergraduate** – Introduction to Economics (lectures, 30 hours, 350–1,000 students); History of Economic Facts and Organisations (lectures, 24 hours).
- **Second-year undergraduate** – Macroeconomics (lectures, 30 hours).
- **Third-year undergraduate** – International Economics (lectures, 30 hours); Territorial and Ecological Transitions (lectures, 12 hours); Sustainable Territorial Policies (lectures, 5 hours).
- **First-year Master's** – Institutional Economics (lectures, 15 hours).
- **Second-year Master's, ETT** – Territorial policies, applied seminars, territorial engineering, supervision of internships and collective projects.

Executive and continuing education

- Second-year Master's GPE (UCA/FERDI) – Monetary Valuation of the Environment (lectures, 10 hours, since 2016).
- Second-year Master's MODEV (UCA/AFD) – Environmental Economics and Microeconomics (lectures, 12 hours, since 2015).
- **Scientific Coordinator** of the IHEDD/FERDI executive training programme *Steering the Ecological*

Transition:

- December 2024 – Tunis, 4 days, in partnership with Expertise France (Savoir Eco project).
- April 2025 – Clermont-Ferrand, 4 days, with I4CE, the Ministry for Ecological Transition, and AFD.
- January 2026 – Clermont-Ferrand, 10 days, with AFD.
- Occasional teaching engagements (FERDI/IHEDD): Rabat, 2022 (UM6P, AFD).

Teaching load

Average over 13 years: **approximately 350 equivalent teaching hours per year** (including approximately 72 hours for administrative responsibilities), with a peak of 502 equivalent teaching hours in 2020–21. Teaching is predominantly lecture-based (70–80 % of total hours) and supported by Moodle course design, including banks of more than 500 self-correcting questions.

Outreach, knowledge transfer, and international activities

- **2024–2025** – Two field missions to Cambodia under the ECOTERM project (2024 kick-off meeting; 2025 co-design workshop with local stakeholders).
- **2025** – Interview for the inaugural issue of UCA's *Déclic* magazine (written and video formats, January 2025).
- **2021–2025** – Member of five steering committees for the *Riom Makes Its Transition* project (sub-prefecture of Puy-de-Dôme; population 20,000): evidence support for public decision-making on sufficiency, resilience, and decarbonisation.
- **2023** – Lecture on planetary boundaries and the ecological transition for staff of the Puy-de-Dôme Prefecture.
- **2018** – IHEDD/FERDI/IDDRI/I4CE training, under the patronage of Laurent Fabius: presentation on implementing the Paris Agreement in the agricultural sector.
- **2017** – One-month visiting appointment at Nanjing Audit University, China: research seminar and two courses delivered.
- **2015** – One-month visiting appointment at the Federal University of Viçosa, Brazil: RAS Symposium, Caraça.

Additional information

- **Languages:** French (native), English (fluent), Portuguese (intermediate).
- **Software and tools:** Stata, R, Matlab, Python (text analysis and web scraping), L^AT_EX, Moodle (course design).